GRADE 7 ACCELERATED · UNIT 5 · LESSONS 2–3

Scale Factor: Perimeter and Area

Question Bank - Additional 5.02 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, find the copy's perimeter or area.

1. Perimeter 26 cm, k = 3.

78 cm

2. Area 9 cm^2, k = 3.

81 cm^2

9 × 9.

3. Area 100 cm^2, k = ½.

25 cm^2

100 × ¼.

4. A 4 cm by 5 cm rectangle, k = 1.5. Perimeter and area of the copy.

P = 27 cm, A = 45 cm^2

18 × 1.5; 20 × 2.25.

For questions 5 through 7, find k, or the ratio asked.

5. Areas 11 cm^2 and 99 cm^2. Find k.

k = 3

99 ÷ 11 = 9 = k^2.

6. Areas 32 cm^2 and 2 cm^2, original to copy. Find k.

k = ¼

2 ÷ 32 = 1⁄16 = k^2, so k = ¼.

7. Perimeters 10 cm and 50 cm, original to copy. What is the area ratio?

25

k = 5; k^2 = 25.

Choose one.

8. A figure with area 4 cm^2 is scaled by k = 6. What is the copy's area?

- ○ A. 48 cm^2
- ● **B. 144 cm^2**
- ○ C. 24 cm^2
- ○ D. 10 cm^2

B

$4 \times 6^2 = 4 \times 36$.